

Motion graphs

A graph is a visual representation of the data of a physical phenomenon. In this case, we are going to represent in graphs, for simple motions, the relationship that exists between:

- Time and position.
- Time and velocity.
- Time and acceleration.

Important rules when plotting graphs:

- They are drawn on a two-dimensional coordinate system:
 - The horizontal axis or X-axis always shows time t . This can be measured in seconds (s), minutes (min), hours (h), or other units of time such as days, years, etc., although the latter are less common.
 - The vertical axis or Y-axis will show the other quantity: position, velocity, or acceleration. Its units must always be indicated:
 - * Position (represents the variable x): has units of length, generally m or km .
 - * Velocity (v): its most common units are $\frac{m}{s}$ or $\frac{km}{h}$.
 - * Acceleration (a): its most common units are $\frac{m}{s^2}$, although sometimes we may see $\frac{km}{h^2}$. In any case, they are always: $\frac{\text{unit of distance}}{\text{unit of time}^2}$.

Types of motion

The motions most commonly studied are:

- **Uniform Rectilinear Motion (URM):**
 - The object moves in a straight line.
 - Its velocity is constant ($v = cte \frac{m}{s}$). This is the same as saying that it has no acceleration or that its acceleration is 0 ($a = 0 \frac{m}{s^2}$).
- **Uniformly Accelerated Rectilinear Motion (UARM):**
 - The object moves in a straight line.

- It has a constant acceleration ($a = cte \frac{m}{s^2}$).
- Constant acceleration means that velocity changes, but always at the same rate. In other words, there is a linear mathematical relationship between velocity, time, and acceleration. The velocity will be higher the longer the acceleration acts (if it is positive): $v = a \cdot t + k$.

Position-Time Graph

Position-time graphs show the position of a moving object (a body in motion) over time.

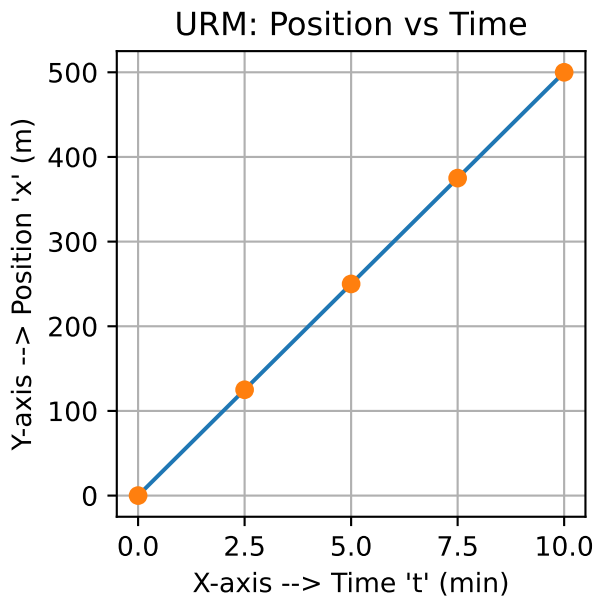
These graphs always have two axes:

- Horizontal axis (*X-axis*), shows time, t , and its units will usually be s , min , or h .
 - It always starts at the origin ($X=0$) and time moves forward to the right. It is as if we started the stopwatch (0) at the origin, and as time passes we move toward the right.
 - **Important: remember that in time we never move backwards, that is, toward the left side of the X-axis.**
- Vertical axis (*Y-axis*), shows the position x . Its units will generally be m or km .
 - The origin ($Y=0$) indicates a fixed position.
 - As we move away from the origin, the position x becomes larger, and on the graph we will see that we move upward as time advances. It will look like a diagonal line rising to the right.
 - If we move away and then return toward the origin, what we will see is a diagonal line descending to the right, but never to the left!, because we can never go back to past times.

Example (outward journey)

The following graph shows how a person leaves their house (which we consider the origin) on the way to school, which is 500 m away.

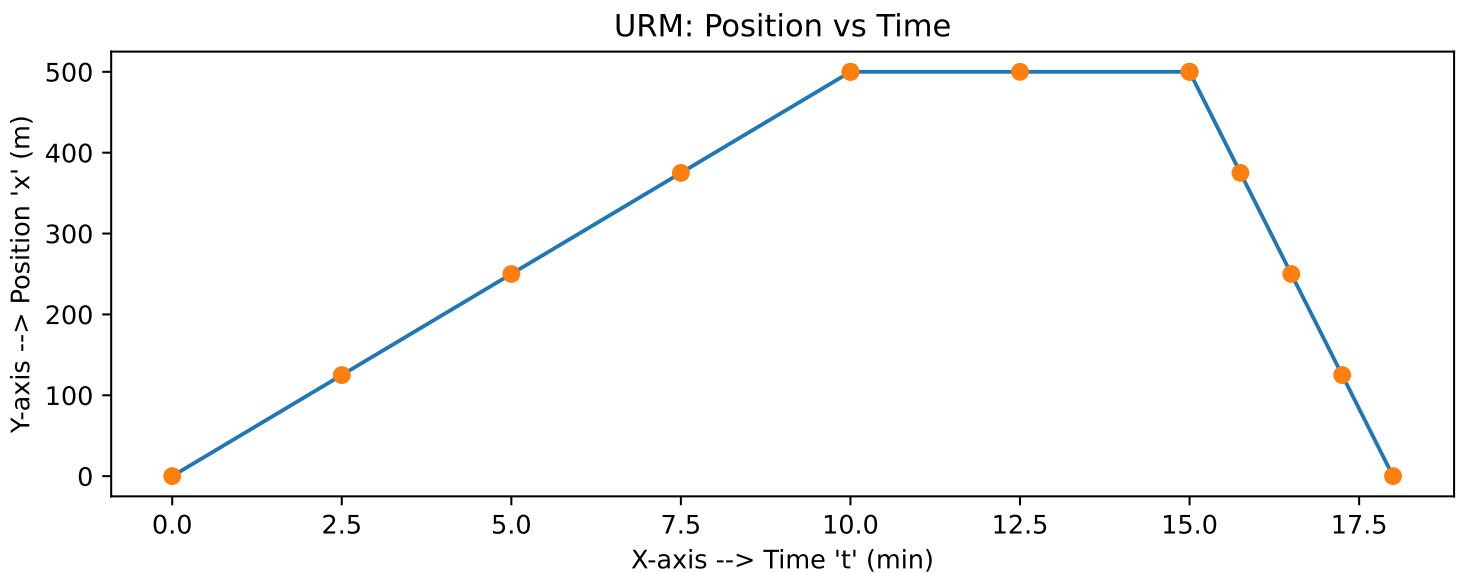
We can see that as time passes (toward the right), their vertical position gets farther and farther away from the X-axis (which corresponds to the position $x = 0m$).



Example (round trip)

In the following graph, we can see how the person goes to school ($x = 500m$), stays there for a few minutes, and then returns home. Do they return at the same speed? No, because we can see that they travel the same distance in less time.

Now we can see that the return trip is shown approaching the horizontal axis ($x = 0m$) with a diagonal line decreasing toward the right.



Question: Looking at the graph, where do you think the person is moving faster: on the way there or on the way back? Why? Explain your reasoning in relation to the velocity formula you know: $v = \frac{d}{t}$.

Ejercicios:

Cyclist Exercise

A cyclist moves in a straight line at a constant speed. At the initial moment $t = 0$ s, they are at position $x = 10$ m. At 5 seconds, they are at position $x = 30$ m.

Required:

1. Calculate the cyclist's speed.
2. Write the equation of motion.
3. Complete the table of values:

t (s)	x (m)
0	
2	
4	
6	
8	

4. Represent:

- Position-time graph $x(t)$
- Velocity-time graph $v(t)$

Solutions

1. Speed

$$v = \frac{30 - 10}{5} = 4 \text{ m/s}$$

2. Equation of motion

$$x(t) = 10 + 4t$$

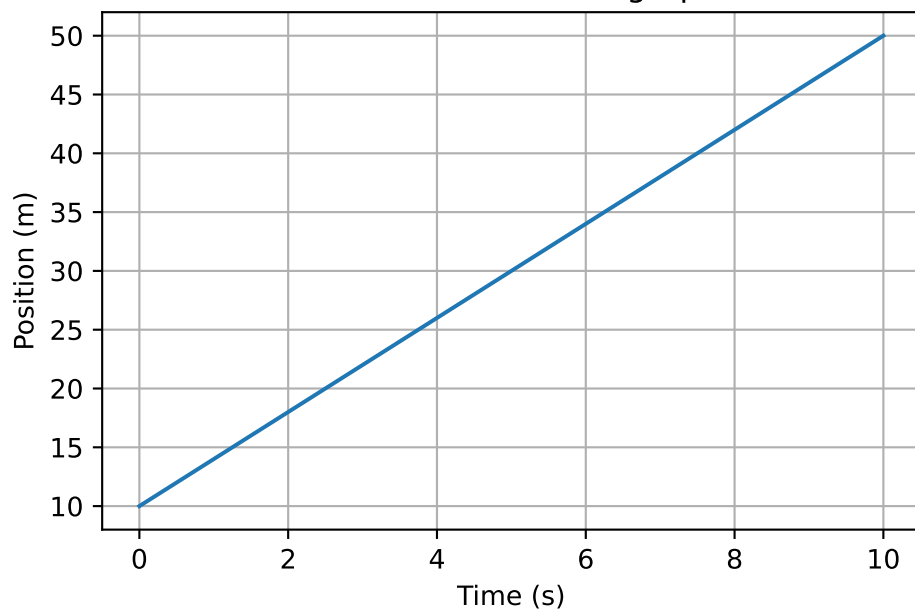
3. Table

t (s)	x (m)
0	10
2	18
4	26
6	34
8	42

4. Graphs

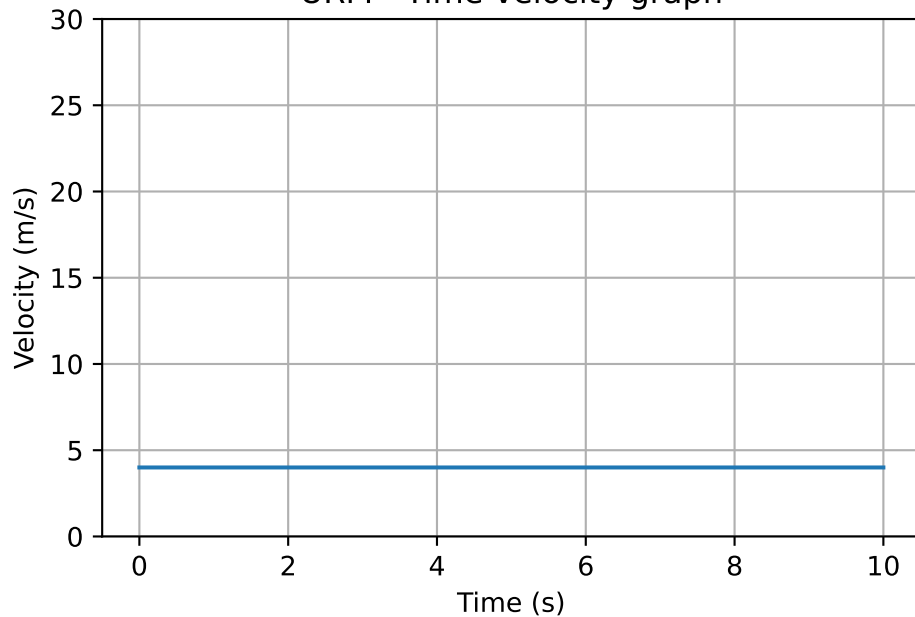
Position-time graph:

URM - Position-time graph



Time-velocity graph

URM - Time-velocity graph



Car Exercise

A car moves at a constant speed of 72 km/h. At the initial moment, it is at $x = -20$ m.

Required:

1. Convert the speed to m/s.
2. Write the equation of motion.

3. Calculate the position at 3 s, 6 s, and 10 s.

4. Represent:

- $x(t)$ graph
- $v(t)$ graph

Solutions

1. Speed conversion

$$72\text{ km/h} = 20\text{ m/s}$$

2. Equation of motion

$$x(t) = -20 + 20t$$

3. Positions

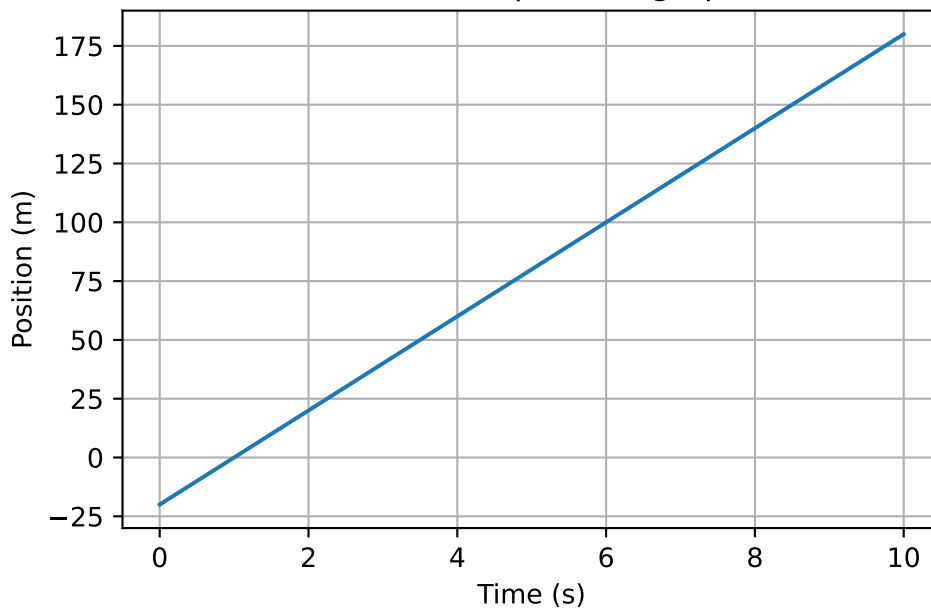
- $x(3) = 40\text{ m}$
- $x(6) = 100\text{ m}$
- $x(10) = 180\text{ m}$

4. Graphs

Position-time graph:

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URM - Time-position graph



Time-velocity graph:

URM - Time-velocity graph

